

M6: Homework

● Graded

Student

Jonathan C Elder

Total Points

46.25 / 50 pts

Question 1

6.1.2

5 / 5 pts

✓ - 0 pts Correct

Question 2

6.1.5

3.75 / 5 pts

✓ - 1.25 pts c incorrect

Question 3

6.2.1

5 / 5 pts

✓ - 0 pts Correct

Question 4

6.2.5

5 / 5 pts

✓ - 0 pts Correct

Question 5

6.3.3

5 / 5 pts

✓ - 0 pts Correct

Question 6

6.4.4

5 / 5 pts

✓ - 0 pts Correct

Question 7

6.5.1

4 / 5 pts

✓ - 1 pt a partially correct

Question 8

6.6.2

3.5 / 5 pts

✓ - 1.5 pts parially correct

Question 9

6.7.4

5 / 5 pts

✓ - 0 pts Correct

Question 10

6.8.1

5 / 5 pts

✓ - 0 pts Correct

Question assigned to the following page: [1](#)

6.1 HW

6.1.2 a) What is the probability that Celia is first in line?

$$\text{probability} = \frac{(n-1)!}{n!} = \frac{(n-1)!}{n(n-1)!} = \frac{1}{n}$$

$$p = \frac{1}{n}$$

b) total way = $n!$
celia = first position
Felicity = last position

$$\text{probability} = \frac{(n-2)!}{n!} = \frac{(n-2)!}{n(n-1)(n-2)!} = \frac{1}{n(n-1)}$$

$$p = \frac{1}{n(n-1)}$$

c) Probability that Celia and Felicity are next to each other?
they act as one object.

They also have 2 total positions they can be in while remaining next to each other.

$$\text{Probability} = \frac{2(n-1)!}{n!} = \frac{2 \times (n-1)!}{n \times (n-1)!} = \frac{2}{n}$$

$$p = \frac{2}{n}$$

Question assigned to the following page: [2](#)

6.1 HW Cont.

- 6.1.5 a) There are four ways of a, b remaining cards selecting three of the same cards.
a, b are such that 3, 3 (meaning there are 6 total ways of getting two of the same pair. $\rightarrow {}^6C_2 = 6$)

Total number of ways 5 cards are such that three are the same

$$= 13 \times 4 \times 6 \times 12$$

$$= 3744$$

$$\text{Probability} = \frac{3744}{{}^{52}C_5} = \frac{3744}{2598960} = 0.00144058$$

chance

- b) 13×4 are the number of ways of getting the same card.
Remaining two cards are selected such that they are of different rank $\rightarrow {}^{49}C_2$

$$\text{Probability} = \frac{(13 \times 4) \times {}^{49}C_2}{{}^{52}C_5}$$

$$= \frac{61152}{2598960} = 0.02352941$$

- c) $4 \times {}^{13}C_5$ are the total number of ways.

$$\text{Probability} = \frac{4 \times {}^{13}C_5}{{}^{52}C_5} = \frac{5148}{2598960} = 0.00198079$$

- d) Probability that two are the same rank & 3 are different?
total ways = $13 \times 6 \times {}^{50}C_3$

$$= 1528800$$

$$\text{Probability} = \frac{1528800}{{}^{52}C_5} = 0.58823529$$

Question assigned to the following page: [3](#)

6.2 HW

6.2.1 a) At least $n-1$ flips come up heads.

Probability all flips come up heads

$$P = \left(\frac{1}{2}\right)^n$$

Probability that $n-1$ comes up heads

$$P = n\left(\frac{1}{2}\right)^n$$

So, Probability that at least $n-1$ flips come up heads is:

$$P = n\left(\frac{1}{2}\right)^n + \left(\frac{1}{2}\right)^n$$

$$= (n+1)\left(\frac{1}{2}\right)^n \rightarrow \boxed{P = (n+1)\left(\frac{1}{2}\right)^n}$$

b) Probability of no two consecutive flips:

$$P = 2\left(\frac{1}{2}\right)^n$$

Probability that at least two consecutive flips are the same:

$$\text{Probability} = 1 - 2\left(\frac{1}{2}\right)^n$$

$$\boxed{P = 1 - \frac{1}{2^{n-1}}}$$

c) n is even, let $n = 2m$

of heads = b

of tails = a

$$P(b \neq a) = 1 - P(b = a)$$

$$= 1 - \binom{n}{n/2} \left(\frac{1}{2}\right)^n$$

$$= 1 - \binom{2m}{m} \left(\frac{1}{2}\right)^{2m}$$

$$= 1 - \frac{(2m)!}{m! m!} \times \left(\frac{1}{2}\right)^n$$

$$\boxed{= 1 - \frac{n!}{\left(\frac{n}{2}\right)! \times \left(\frac{n}{2}\right)!} \times \left(\frac{1}{2}\right)^n}$$

Question assigned to the following page: [4](#)

6.2 HW Cont.

6.2.5 Number of digits = 10
Number of uppercase letters = 26
Number of lowercase letters = 26
Number of characters for password = 10

$$\text{Total possible strings} = (26 + 26 + 10)^{10} \\ = 62^{10}$$

Password-w/ 10 strings that have at least 1 uppercase,
1 lowercase, and 1 digit?

$$= 26 \times 26 \times 10 \times 62^7$$

$$= 23806114737966080$$

Probability = required outcome / total possible strings

$$P = \frac{23806114737966080}{62^{10}}$$

$$P = 0.0283642711$$

Question assigned to the following page: [5](#)

6.3 HW

6.3.3 a) total people = 8

total arrangements = $8!$

consider the bride and groom as 1 person, so there are then 7 people

Total arrangements = $7!$

Bride/Groom arrangements = 2 ways

$$\text{Probability} = \frac{2 \times 7!}{8!} = \frac{2 \times 7!}{8 \times 7!} = \frac{2}{8} = \frac{1}{4}$$

b) There is only 1 way to line up the maid of honor to the leftmost.

Then the remaining can be assigned in $7!$ ways

$$\text{Hence, Probability} = \frac{7!}{8!} = \frac{7!}{8 \times 7!} = \frac{1}{8}$$

c) False

They have dependent events that include an intersection.

The condition in part a has $6! \times 2$ ways that are included in the intersection.

Question assigned to the following page: [6](#)

6.4 HW

6.4.4 $P(D)$ = probability of testing positive for HIV

$P(T)$ = probability of the test

$$P(D|T) = \frac{P(T|D) \cdot P(D)}{P(T|D) \cdot P(D) + P(T|ND) \cdot P(ND)}$$

$P(ND)$ = probability of not having HIV

$$P(D) = 0.0001$$

$$P(ND) = 0.9999$$

$$P(T|D) = 1$$

$$P(T|ND) = 0.025$$

$$P(D|T) = \frac{1(0.0001)}{1(0.0001) + (0.025)(0.9999)}$$

$$= 0.0039$$

Question assigned to the following page: [7](#)

6.5 HW

6.5.1 When 2 dice are thrown, these are all outcomes:

$6^2 = 36$

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

favorable cases: $\{(1,6), (2,3), (3,2), (6,1)\} = 4$

$$P(X=6) = \frac{\text{favorable cases}}{\text{total cases}} = \frac{4}{36} = \frac{1}{9}$$

$$P(X=6) = \frac{1}{9}$$

Question assigned to the following page: [8](#)

6.6 HW

6.6.2

G	P(G=F)
-1	4/6 = 2/3
1	1/6
2	1/6

$$\text{Expected value} = E(x) = (-1) \times \frac{2}{3} + 1 \times \left(\frac{1}{6}\right) + 1 \times \left(\frac{1}{6}\right) = -\frac{1}{3}$$

$$E(x^2) = (-1)^2 \times \frac{2}{3} + 1^2 \times \left(\frac{1}{6}\right) + 1^2 \times \left(\frac{1}{6}\right) = 1$$
$$E(x^2) - (E(x))^2 = 1 - \left(-\frac{1}{3}\right)^2 = 1 - \left(\frac{1}{9}\right) = \frac{8}{9} \text{ or } .889$$

Question assigned to the following page: [9](#)

6.7 HW

6.7.4 Let X = number of student who get their own coat.

$$\text{To find } E(X) = \sum x P(X=x)$$

$$X_i = \begin{cases} 1, & \text{if they select their own coat} \\ 0, & \text{if they do not select their own coat.} \end{cases}$$

You can then expect the number of students to get their own coat to be the sum of these indicators such that:

$$X = X_1 + X_2 + \dots + X_{10}$$
$$E(X) = E(X_1) + E(X_2) + \dots + E(X_{10})$$

$$\text{Then, } E(X) = 1 \times \frac{1}{10} + 1 \times \frac{1}{10} + \dots + 1 \times \frac{1}{10}$$

$$= 10 \times \frac{1}{10} = \text{1 child is expected to grab their own coat.}$$

Question assigned to the following page: [10](#)

6.8 HW

6.8.1

a) $n=100$

$$P=1\% = \frac{1}{100} = 0.01$$

x = number of defects

$$P(X=k) = \binom{n}{k} P^k (1-P)^{n-k}$$

$$P(X=2) = \binom{100}{2} P^2 (1-P)^{100-2}$$

$$P(X=2) = \binom{100}{2} (0.0001) (1-0.01)^{100-2}$$

$$P(X=2) = 0.1849$$

$$b) P(X=0) + P(X=1) = \binom{100}{0} (0.0001)^0 (1-0.01)^{100-0} + \binom{100}{1} (0.0001)^1 (1-0.01)^{100-1}$$

$$1 \text{ defect} = .7358$$

$$P(X \geq 2) = 1 - [P(X=0) + P(X=1)]$$

$$= 1 - 0.7358$$

$$P(X \geq 2) = 0.2642$$

c) Expected number of defects $E(X) = nP$

$$P = 1\% = 0.01$$

$$n = 100$$

$$E(X) = 100 \times 0.01 = 1 \text{ defect}$$

d) For batches $n = 50$

$$P = 1\% = .01 \text{ and if } X \sim B(n, P)$$

$$P(X \geq 1) = 1 - P(X=0)$$

$$= 1 - 0.605$$

$$= .395$$

$$P(X=0) = \binom{50}{0} \left(\frac{1}{100}\right)^0 \left(\frac{99}{100}\right)^{50-0}$$

$$= 1 \times 1 \times (.99)^{50}$$

Expected number of defective = .605

$$\text{batches} = np = 50 \times .01 = .5$$

Each batch has 2 defects

Expected number of defective boards = 1